Year 11 Set 4 · Foundation · Lesson 3

Solving linear equations

w/c 9 November 2026 · OnMaths · Created by Mr Ali

Today I will learn to...

Goals

1. Solve two-step equations and equations with brackets

2. Solve equations with unknowns on both sides

Do now: retrieval practice

4 mins

1. Share £50 in the ratio 2 : 3

   £20 and £30

2. Simplify 3x + 2x − x

   4x

3. Expand 4(x + 3)

   4x + 12

4. Work out −3 × −5

   15

Objective 1

Solve two-step equations and equations with brackets

Key idea

Learn

!

Undo operations in reverse order. Whatever you do to one side, do to the other.

Objective 1: Solve two-step equations and equations with brackets

I do: example 1

I do

Solve $3x + 5 = 20$

1. $3x + 5 = 20$
2. Subtract 5 from both sides: $3x = 15$
3. Divide both sides by 3: $x = 15 \div 3$
4. $x = 5$ (check: $3 \times 5 + 5 = 20$ ✓)

I do: example 2

I do

Solve $4(x - 2) = 24$

1. $4(x - 2) = 24$
2. Divide both sides by 4: $x - 2 = 6$
3. Add 2 to both sides: $x = 6 + 2$
4. $x = 8$ (check: $4 \times 6 = 24$ ✓)

I do: example 3

I do

Solve x/3 + 4 = 10

1. x/3 + 4 = 10
2. Subtract 4 from both sides: x/3 = 6
3. Multiply both sides by 3: x = 6 × 3
4. x = 18 (check: 18 ÷ 3 + 4 = 10 ✓)

Mini whiteboards

We do

1. Solve $2x - 7 = 11$

Answer: $x = 9$

2. Solve $5(x + 1) = 35$

Answer: $x = 6$

3. Solve $3x + 4 = 19$

Answer: $x = 5$

4. Solve $4(x - 2) = 20$

Answer: $x = 7$

You do: choose your challenge

You do

Bronze

Solve $5x + 2 = 27$

Silver

Solve $3(x - 4) = 18$

Gold

I think of a number, multiply it by 4 and subtract 7. The answer is 29. What was my number?

Platinum

Solve $(2x + 5)/3 = 7$

You do: answers

Check

Bronze	x = 5
Silver	x = 10
Gold	4n − 7 = 29, n = 9
Platinum	2x + 5 = 21, x = 8

If you finish early

Extra

Bronze

Solve $2x + 9 = 21$

$x = 6$

Silver

Solve $6(x + 3) = 48$

$x = 5$

Gold

Solve $(3x - 1)/4 = 5$

$3x - 1 = 20$, $x = 7$

Platinum

Solve $5(2x - 3) = 3(x + 2)$

$10x - 15 = 3x + 6$, $x = 3$

Exam question 1

2 marks

Solve $6x - 4 = 29$

Space for working

Mark scheme: $6x = 33$, $x = 5.5$

Exam question 2

3 marks

Solve $3(2x + 1) = 27$

Space for working

Mark scheme: $2x + 1 = 9$, $x = 4$

Objective 1 achieved

Solve two-step equations and equations with brackets



Objective 2

Solve equations with unknowns on both sides

✓ Achieved: Solve two-step equations and equations with brackets

Key idea

Learn

!

Move the smaller x-term first so the x-term stays positive, then solve as normal.

Objective 2: Solve equations with unknowns on both sides

I do: example 1

I do

Solve 5x + 3 = 2x + 18

1. $3x + 3 = 18$
2. $3x = 15$
3. $x = 5$

I do: example 2

I do

Solve $7x - 4 = 3x + 10$

1. $4x - 4 = 10$
2. $4x = 14$
3. $x = 3.5$

I do: example 3

I do

Solve 2(x + 5) = x + 16

1. $2x + 10 = x + 16$
2. $x + 10 = 16$
3. $x = 6$

Mini whiteboards

We do

1. Solve $4x + 1 = x + 13$

Answer: $x = 4$

2. Solve $6x - 5 = 2x + 7$

Answer: $x = 3$

3. Solve $7x - 2 = 3x + 10$

Answer: $x = 3$

4. Solve $2x + 9 = 5x$

Answer: $x = 3$

You do: choose your challenge

You do

Bronze

Solve $3x + 4 = x + 12$

Silver

Solve $5x - 3 = 2x + 9$

Gold

Sam has $3x + 5$ sweets and Ben has $x + 17$.
They have the same number. Find x.

Platinum

Solve $4(x + 3) = 2(x + 9)$

You do: answers

Check

Bronze

$x = 4$

Silver

$x = 4$

Gold

$2x = 12$, $x = 6$

Platinum

$4x + 12 = 2x + 18$, $x = 3$

If you finish early

Extra

Bronze

Solve $5x + 2 = 3x + 10$

$x = 4$

Silver

Solve $8x - 7 = 3x + 18$

$x = 5$

Gold

Solve $3(x + 4) = x + 20$

$x = 4$

Platinum

Solve $2(3x - 1) = 4(x + 2)$

$6x - 2 = 4x + 8$, $x = 5$

Exam question 1

3 marks

Solve $8x - 3 = 5x + 12$

Space for working

Mark scheme: $3x = 15$, $x = 5$

Exam question 2

3 marks

A rectangle has sides (x + 4) cm and (2x − 1) cm. Its perimeter is 30 cm. Find x.

Space for working

Mark scheme: 2(3x + 3) = 30 → 6x + 6 = 30 → x = 4

Objective 2 achieved

Solve equations with unknowns on both sides



Real life

Apply

A taxi charges £3 plus £2 per mile. A journey costs £19. How many miles was it?

3 + 2m = 19 → m = 8 miles

Extension: stretch yourself

Stretch

1. Solve $(x + 3)/2 = (x - 1)/3$

$3x + 9 = 2x - 2 \rightarrow x = -11$

2. The angles of a triangle are x, 2x and 3x + 30. Find x.

$6x + 30 = 180 \rightarrow x = 25$

Exit ticket: exam questions

Exit

1. Solve $7x + 3 = 38$ **(2 marks)**

2. Solve $2(3x - 1) = 22$ **(3 marks)**

3. Solve $9x - 5 = 4x + 20$ **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 $x = 5$ [2]

Q2 $6x - 2 = 22$, $x = 4$ [3]

Q3 $5x = 25$, $x = 5$ [3]

Total: ____ / 8

Today I achieved

- ✔ **Solve two-step equations and equations with brackets**
- ✔ **Solve equations with unknowns on both sides**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Solve $4x - 9 = 15$ [2]

2. Solve $x/5 + 3 = 7$ [2]

3. Solve $3(2x + 4) = 30$ [3]

4. Solve $6x + 1 = 2x + 19$ [3]

5. The angles in a quadrilateral are x, $x + 30$, $2x$ and $2x + 30$. Find x. [3]

Extra practice: Maths Genie → Edexcel question bank → Solving Equations